MODULE 08 · DOMAIN 2 - CLAUDE MODELS, PROMPTING AND CONTEXT ENGINEERING

System Prompts, Templates and Techniques

Designing production prompts, choosing zero-shot, few-shot or thinking, and knowing what a prompt can and cannot guarantee.

DOMAIN 2 of 7 EXAM WEIGHT 13% MODULE 8 of 38

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A system prompt is a briefing for a brilliant new colleague who **missed every meeting**.

Anatomy of a production system prompt

Examples
Three to five
Tagged, varied

Role
One sentence
Focuses behaviour

Tool guidance
When to use each
Plain wording

Context and why
Audience, purpose
Reasons for the rules

System prompt
Stable across requests
Variable input comes last
Versioned, tested, cached

Scope and rules
In and out of scope
Fixed refusal wording

Output format
Structure or schema
Length and tone

THE TAKEAWAY

Stable parts first and cached; the request's own input last.

Write at the right altitude

TOO LOW

Hard-coded logic

- Dozens of if-this-then-that rules
- Brittle on cases nobody listed
- Hard to maintain

TOO HIGH

Vague wishes

- "Be helpful and professional"
- Assumes shared context
- Nothing to test against

THE TIEBREAKER

The right altitude: clear principles and heuristics, with the reasons, plus a few concrete examples.

Templates: fixed parts and variable slots

PART OF THE PROMPT	FIXED OR VARIABLE
Role, rules, format, examples	Fixed: versioned, cached prefix
Reference documents	Fixed if stable; retrieved if not
Retrieved passages for this request	Variable
Customer or user message	Variable, placed last

ON THE EXAM

Templates make prompts testable: change one version, run the evaluation set, compare.

Three techniques, three triggers

TECHNIQUE	REACH FOR IT WHEN
Zero-shot	The task and categories are clearly defined in the prompt
Few-shot	Format, tone or edge cases must be consistent: 3-5 tagged examples
Thinking	The problem needs multi-step reasoning, worth the extra latency

ON THE EXAM

Anthropic: thinking adds latency, so use it when it meaningfully improves quality. When in doubt, respond directly.

Zero-shot or few-shot?

ZERO-SHOT

Definitions do the work

- Clear categories with descriptions
- Output format stated
- Fewer tokens, easy to maintain

FEW-SHOT

Examples do the work

- Exact format or tone matters
- Edge cases keep going wrong
- Relevant, diverse, tagged

THE TIEBREAKER

Start zero-shot. Add examples when evaluation shows format or edge-case errors, not by default.

Instructions that work, from Anthropic's guidance

- **Be explicit, and say why.** Claude generalises from the reason.
- **Say what to do,** not only what to avoid.
- **Separate parts with XML tags:** instructions, documents, examples, input.
- **Long documents at the top, the question at the end:** up to 30% better.
- **Drop the shouting.** "CRITICAL: you MUST" can over-trigger on newer models.

> ON THE EXAM
>
> Clear, calm, structured and reasoned beats loud and long.

Contradictions: resolve them at the source

TWO RULES THAT COLLIDE

Section 1 and section 7

```
1 Answer only from the policy
2 documents supplied.
3 ...
4 Never leave the adviser without
5 an answer. Fill any gaps with
6 your best judgement.
```

Claude has to guess which rule wins, case by case.

ONE RULE, ONE FALLBACK

Rewritten

```
1 Answer only from the policy
2 documents supplied.
3 If they do not answer the question,
4 say what is missing and offer to
5 transfer the call to a specialist.
```

Grounded, with a defined path when sources run out.

THE TIEBREAKER

Never ask the model to arbitrate between your own rules. Remove the conflict and define the fallback.

Guardrails in the prompt: necessary, not sufficient

- **Define scope:** what is in, what is out, and the exact refusal wording.
- **Treat retrieved and user content as data,** never as instructions.
- **A prompt lowers the odds;** it does not enforce anything.
- **Pair it with controls outside the model:** classifiers, validation, permissions.

> ON THE EXAM
>
> A prompt guardrail is the first layer of defence in depth. Module 22 builds the rest.

Outputs for machines, and prompts across versions

- **Machine-read output:** use structured outputs, not a prefilled reply.
- **Prefill is rejected from Opus 4.6;** structured outputs replace it.
- **Techniques are measured per model.** Re-check them on your evals after an upgrade.
- **Newer models follow system prompts more closely,** and may format less by default.

> ON THE EXAM
>
> A prompt is tuned to a model. Upgrade the model, re-run the prompt's evaluation.

> Mereside's complaints team has defined 14 complaint categories, each with a clear written description. Every complaint must be assigned one category code for routing.
>
> Corvane is choosing the starting prompt technique.

Which starting point fits best?

A. A zero-shot prompt with the category descriptions and the required output format, validated on a test set

B. A long instruction to reason step by step before every label

C. Fifty labelled examples covering every category, from day one

D. An agent that reads each complainant's account history before classifying

◆ PAUSE AND DECIDE

Which starting point fits best?

A. A zero-shot prompt with the category descriptions and the required output format, validated on a test set

Clear definitions do the work. Start simple, then measure.

B. A long instruction to reason step by step before every label

Latency and tokens on a single-label decision, by default.

C. Fifty labelled examples covering every category, from day one

Examples are for errors the evaluation shows, not a starting tax.

D. An agent that reads each complainant's account history before classifying

Climbs the ladder for a single classification.

THE SIGNAL

Well-defined categories with descriptions: start zero-shot, and let evaluation decide what to add.

> Quillon's prior-authorisation letters vary in structure: some cite the insurer's criteria by number, some paraphrase them, and section order changes from letter to letter.
>
> The clinical team has approved four exemplary letters, covering different insurers and case types.

Which change will most improve consistency?

A Raise the effort setting on the drafting call

B Add the approved letters as tagged examples, varied across insurers and case types

C Add one example letter and instruct Claude to copy it exactly

D Move the drafting to the most capable tier

◆ PAUSE AND DECIDE

Which change will most improve consistency?

A. Raise the effort setting on the drafting call

More reasoning, but nothing shows what the house structure is.

B. Add the approved letters as tagged examples, varied across insurers and case types

Few-shot: relevant, diverse, tagged examples set the format.

C. Add one example letter and instruct Claude to copy it exactly

One example gets over-copied, including details that should vary.

D. Move the drafting to the most capable tier

A stronger model still has not been shown the format.

THE SIGNAL

Inconsistent format plus approved exemplars: few-shot, with diverse tagged examples.

> Tallyhawk's support router identifies the intent of short customer messages, such as "reset my password" or "change my billing email". It is latency-sensitive.
>
> An engineer added "think step by step before answering" to every prompt. Latency roughly doubled, and routing accuracy did not change.

What should Corvane recommend?

A Keep the instruction, because reasoning always improves quality

B Raise effort so the reasoning is more thorough

C Expand the instruction into a ten-step reasoning procedure

D Remove it from intent routing, and keep reasoning for multi-step diagnosis routes where evals show it helps

◆ PAUSE AND DECIDE

What should Corvane recommend?

A Keep the instruction, because reasoning always improves quality

The measurement says otherwise: same accuracy, double the latency.

B Raise effort so the reasoning is more thorough

Even slower, on a task that does not need reasoning.

C Expand the instruction into a ten-step reasoning procedure

Prescriptive steps, more tokens, still no gain.

D Remove it from intent routing, and keep reasoning for multi-step diagnosis routes where evals show it helps

Reasoning where it pays; direct answers where it does not.

THE SIGNAL

Doubled latency with no accuracy gain: chain of thought was the wrong default for this task.

> Mereside's agent-assist prompt says in section 1: answer only from the policy documents supplied. Section 7, added later, says: never leave an adviser without an answer; use your best judgement to fill gaps.
>
> Answers to the same question sometimes cite policy and sometimes improvise.

What is the most appropriate fix?

A Add a line telling Claude to apply whichever instruction best fits each question

B Remove the gap-filling rule and define a fallback: say what is missing and offer a transfer to a specialist

C Repeat the grounding rule in capital letters at the end of the prompt

D Lower the temperature so answers are more consistent

◆ PAUSE AND DECIDE

What is the most appropriate fix?

A. Add a line telling Claude to apply whichever instruction best fits each question

Hands the arbitration to the model. Inconsistency becomes official.

B. Remove the gap-filling rule and define a fallback: say what is missing and offer a transfer to a specialist

Resolves the conflict at the source, with a safe path when sources run out.

C. Repeat the grounding rule in capital letters at the end of the prompt

Both rules still exist. Shouting may over-trigger.

D. Lower the temperature so answers are more consistent

Consistent wording, same contradiction. Rejected on current Opus anyway.

THE SIGNAL

Inconsistent behaviour from contradictory instructions: fix the prompt, not the model's judgement.

> Quillon's patient-portal prompt says the assistant must never give clinical advice, and gives a fixed refusal pointing to the nurse line.
>
> In adversarial testing, about 2% of cleverly phrased requests still produce clinical advice.

What should Corvane add?

A. Keep the prompt rule, and add an output check that blocks clinical advice and routes the patient to the nurse line

B. Rewrite the rule in capital letters and repeat it three times

C. Keep the prompt as it is and add a disclaimer under every reply

D. Add more examples of polite conversation to the prompt

◆ PAUSE AND DECIDE

What should Corvane add?

A. Keep the prompt rule, and add an output check that blocks clinical advice and routes the patient to the nurse line
Two layers: the prompt lowers the odds; the check enforces.

B. Rewrite the rule in capital letters and repeat it three times
Still only a prompt, and shouting can over-trigger.

C. Keep the prompt as it is and add a disclaimer under every reply
The advice still reaches the patient.

D. Add more examples of polite conversation to the prompt
Unrelated to the leak.

THE SIGNAL

A prompt guardrail that leaks under pressure: keep it, and add enforcement outside the model.

> Mereside's agent-assist prompt starts with the adviser's question, then mixes instructions into the middle of the 70-page lending policy, and ends with the call transcript.
>
> Answers on complex, multi-section questions are weaker than expected.

Which two changes are most likely to improve answers? Choose two.

- A Move the adviser's question to the very start of the prompt
- B Place the policy near the top, with the adviser's question at the end
- C Put CRITICAL in front of every rule
- D Separate instructions, policy and transcript with consistent tags
- E Delete the role sentence to save tokens

◆ PAUSE AND DECIDE

Which two changes are most likely to improve answers? Choose two.

A Move the adviser's question to the very start of the prompt

Already at the start. Anthropic advises the query at the end.

B Place the policy near the top, with the adviser's question at the end

Long documents first, query last: up to 30% better on complex inputs.

C Put CRITICAL in front of every rule

Shouting over-triggers on newer models.

D Separate instructions, policy and transcript with consistent tags

Stops instructions blurring into reference text.

E Delete the role sentence to save tokens

Saves almost nothing, and the role does help.

THE SIGNAL

Long material first, question last, parts clearly tagged.

In five lines

- **Brief a brilliant newcomer:** role, context and why, scope, format, examples, tool guidance.
- **Templates:** fixed parts first and cached, variable input last, versions tested.
- **Zero-shot first; few-shot** for format and edge cases; **thinking** only where it pays.
- **Resolve contradictions at the source.** Never ask the model to arbitrate your rules.
- **Prompt guardrails lower the odds;** enforcement lives outside the model.

COMING UP

Next: context windows and token budgets.